TCGA-BRCA: Gene Expression and PAM50 Visualization

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Introduction

Breast cancer represents a major global public health dilemma requiring multinational synergy in research, and pharmaceutical development. In 2024, approximately 2.4 million women were diagnosed with breast cancer world-wide, with an estimated 694,000 associated deaths(World Health Organization, 2026). Despite advances in screening, and treatment, breast cancer remains a biologically heterogeneous disease, with considerable variable in molecular characteristic and clinical behavior.

Through the contribution of scientists globally, gene-expression research has identified distinct molecular subtypes, Luminal A, Luminal B, Her-2 enriched, and Basal-like breast cancers (Parker et al., 2009). The classifications demonstrate differences in tumor biology and give researcher a frame work for investigating variance in gene expressions. The Cancer Genome Atlas (TCGA), has established an extensive molecular database for investigating breast cancer heterogeneity (Cancer Genome Atlas Network, 2012).

Methods

We examine the supplied gene-expression matrix (`counts.csv`) and associated annotations (`meta_data.csv`). The results shown here are in whole or part based upon data generated by the TCGA Research Network: TGCA. The TCGA is made possible by the selfless contributions of patients who consented to the use of their original biological specimens, and associated clinical information for cancer research. Without donated tissue samples, alongside relevant clinical data, researchers would not be able to make the remarkable advancements in genomic and molecular definement of cancer charateristics. TCGA established informed consent procedures alongside data access safegaurds to protect patients rights, while supporting the needs of ongoing research for the scientific community. The patients contributions to society continue to

advance our ability to analyize and understand breast cancer hetergeneity

[State the verified data origin and assay, how sample identifiers were matched, the selected sample types, exclusions, and the exact subtype field.] [Describe any gene filtering, normalization or transformation, z-score calculation, and how genes were selected for visualization.] Descriptive summaries and a heatmap characterize the selected cohort. An illustrative volcano plot displays a specified between-group contrast; [state the actual comparison, computation of effect sizes and p-values, and any adjustment]. Unless a suitable count-based model was fitted, the contrast is descriptive rather than confirmatory (Benjamini & Hochberg, 1995; Love et al., 2014).

Results

[Report the final matched sample count and the molecular subtype distribution, including denominators and missing annotations.] [Report only reproducible descriptive findings from the expression summaries.] [Insert the final heatmap and briefly describe its most relevant visible patterns, noting that z-score tails are not significance tests.] [Insert the final volcano plot and identify the groups, axes, thresholds, and limitations; do not interpret illustrative highlights as validated differential expression.]

Figures

AI Disclosure

AI was used to type in the file names for the figures. The main file before modifications was cleaned by AI from an old manuscript to empty the file, and pre-cook with reporting basic instructions - NO AI USED FOR MANUAL TYPE UP. After template was cooked, this was hand written, and typed with the exception of the file names. VIEW GITHUB for documentation chain.

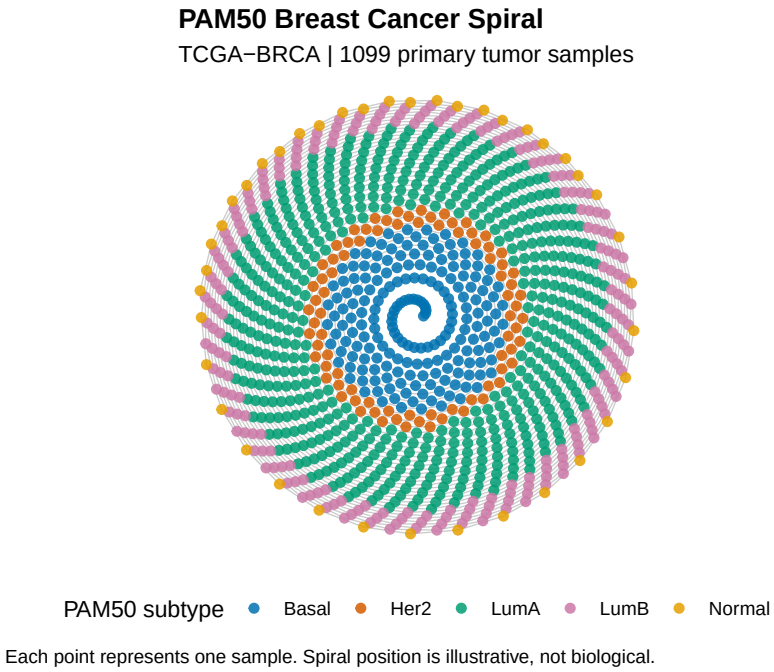


Figure 1: Visualization of PAM50 molecular subtypes in the TCGA-BRCA cohort.

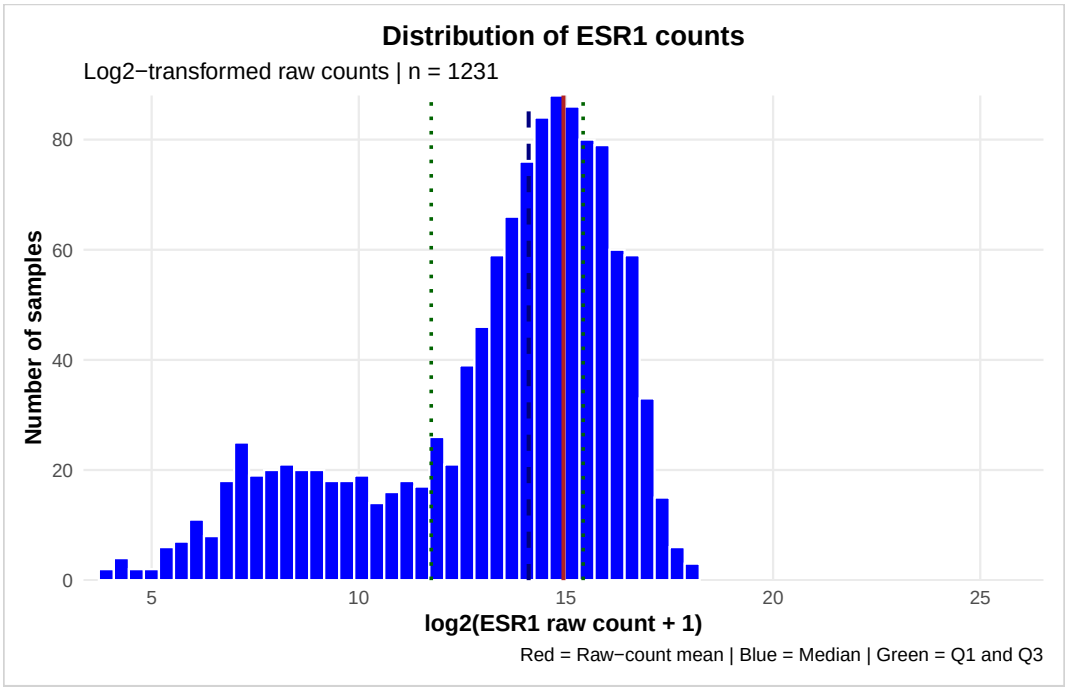


Figure 2: Distribution of ESR1 gene-expression values.

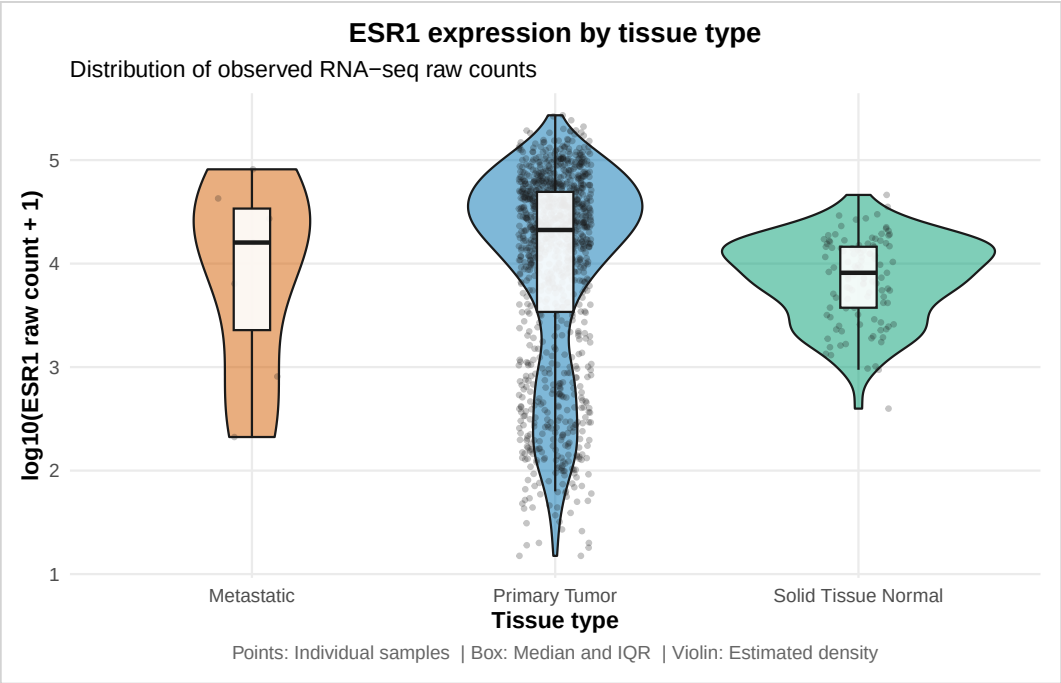


Figure 3: ESR1 expression across sample types.

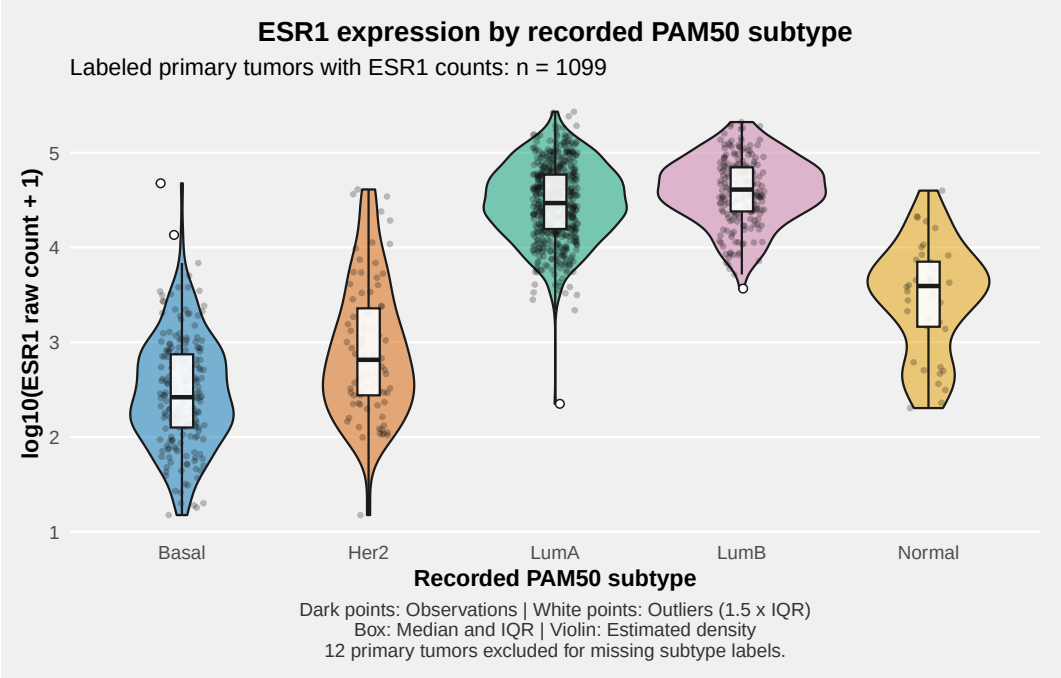


Figure 4: ESR1 expression across PAM50 molecular subtypes.

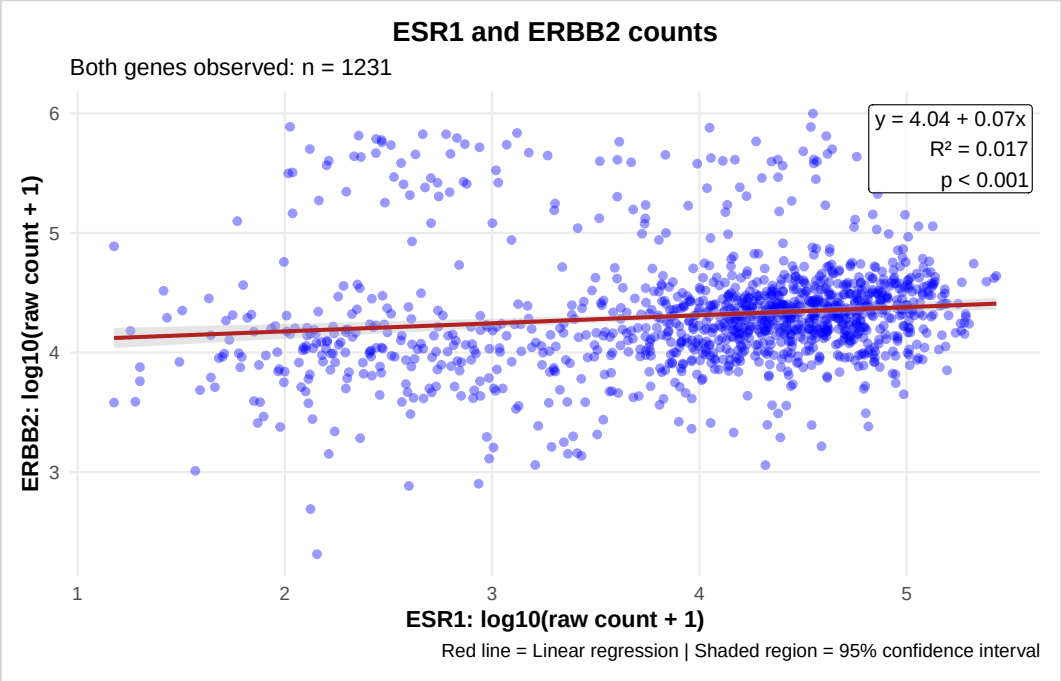


Figure 5: Exploratory gene-expression scatterplot.

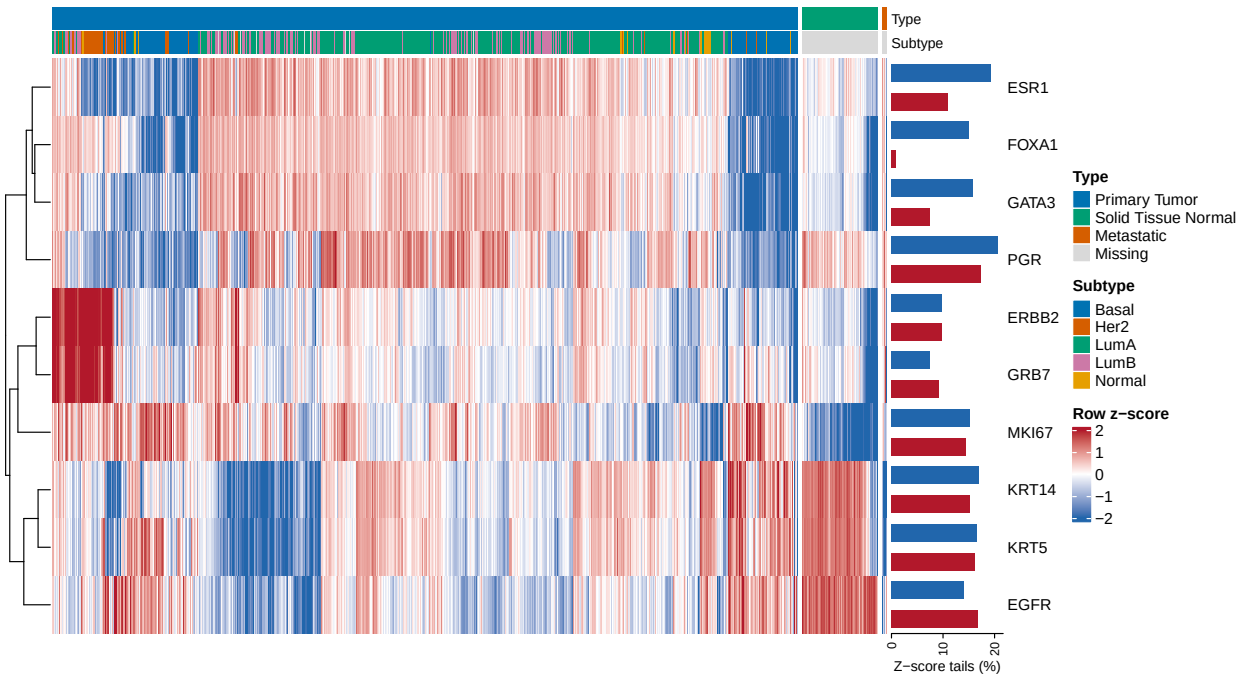


Figure 6: Heatmap of selected breast cancer-associated gene-expression patterns.

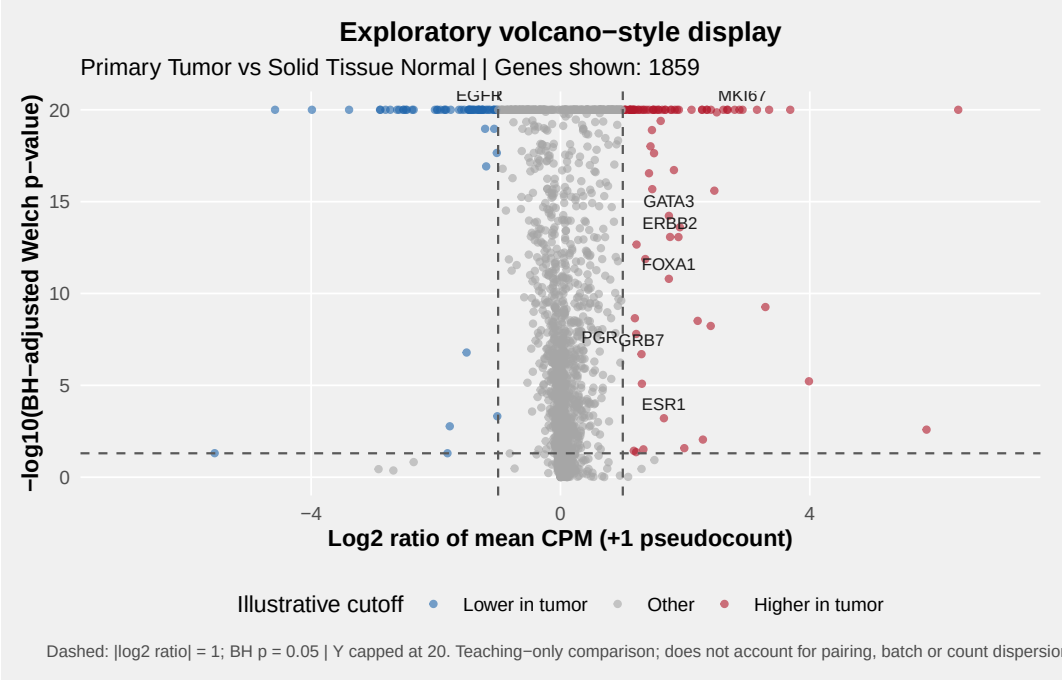


Figure 7: Exploratory volcano plot of gene-expression differences between comparison groups.

References

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Cancer Genome Atlas Network. (2012). Comprehensive molecular portraits of human breast tumours. *Nature*, *490*(7418), 61–70. https://doi.org/10.1038/nature11412

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